Are We Mapping Ecosystems or Models? How Reconstruction Choices Shape Food Web Inference

Tanya Strydom [1*], **Baran Karapunar** [2], **Jennifer A. Dunne** [3], **Pincelli M. Hull** [5], **Crispin T.S. Little** [2], **Catalina Pimiento** [6,7], **Andy Ridgwell** [4], **Andrew P. Beckerman** [1], **Alexander M. Dunhill** [2]

[1] School of Biosciences, University of Sheffield, Sheffield, UK, S10 2TN
[2] School of Earth, Environment and Sustainability, University of Leeds, Leeds, UK, LS2 9JT
[3] Santa Fe Institute, 1399 Hyde Park Road, Santa Fe, NM 87501, USA
[4] Department of Earth and Planetary Sciences, University of California, Riverside, CA, USA
[5] Department of Earth & Planetary Sciences, Yale University, New Haven, CT, USA
[6] Department of Paleontology, University of Zurich, Zurich, Switzerland
[7] Biosciences Department, Swansea University, Swansea SA28PP, UK

[*]Correspondence: t.strydom@sheffield.ac.uk

Abstract: 1. Ecological networks are increasingly used to compare community organisation, quantify interaction turnover, and infer responses to environmental change. However, interactions are often incompletely observed and must be reconstructed using models that encode different ecological assumptions. The extent to which these choices shape ecological inference remains underexplored.

2. We examine this problem by reconstructing four successive marine assemblages spanning the Early Toarcian extinction and recovery in the Cleveland Basin, UK. We compare six food-web models spanning feasibility-based, realised, and structural representations, holding species composition constant within each assemblage. We quantify network topology, pairwise interaction turnover, and inferred extinction dynamics.

3. Model choice explained substantially more variation in network structure than temporal turnover across extinction and recovery phases. Models producing similar network-level properties nevertheless inferred substantially different species-level interactions. Extinction scenario rankings were broadly consistent at the species level, but interaction-level outcomes and cascade dynamics were highly model contingent.

4. These findings demonstrate that the reliability of network-based inference depends on the scale of the ecological question. Similar network structures do not guarantee agreement on individual interactions, and apparently robust aggregate patterns may coexist with substantial uncertainty in inferred mechanisms. Network reconstruction should therefore be treated as an explicit component of hypothesis specification, with approaches matched to inferential goals and conclusions tested across alternative network representations.

Keywords: Network reconstruction, Macroecology, Ecological networks, Interaction turnover, Trophic organisation, Palaeoecology, Ecosystem resilience

1 Introduction

Understanding how biological communities are organised and how species interact with each other is a central goal of ecology. While early efforts focused primarily on species richness and composition, there is growing recognition that ecological communities are structured not only by which species occur, but by how they interact (Gravel *et al.*, 2019; Thuiller *et al.*, 2024; Mestre & Bastazini, 2026). Interaction networks are increasingly treated as macroecological state variables where they are used to compare community organisation across environmental gradients (Tylianakis & Morris, 2017), quantify β-diversity in interaction structure (Poisot *et al.*, 2015; Trøjelsgaard & Olesen, 2016), evaluate stability–complexity relationships (Allesina & Tang, 2012), and infer vulnerability under global change (Hao *et al.*, 2025).

As a result, ecological networks now play a central role in comparative analyses spanning latitudinal gradients, disturbance regimes, and deep time environmental transitions (Roopnarine, 2006; Poisot & Gravel, 2014; Michalska-Smith & Allesina, 2019; Dunhill *et al.*, 2024; Hao *et al.*, 2025). Implicit in this expansion is the critical assumption that network properties estimated across systems using various models are structurally comparable, and that differences among network properties reflect ecological signal rather than methodological artefact (Fründ *et al.*, 2016; Jordano, 2016). However, most ecological networks are not fully observed as interaction data are incomplete and sampling is uneven across historical and biogeographic contexts, across both present day and deep time (Poisot *et al.*, 2021; Sandra *et al.*, 2025).

Interactions must often be inferred indirectly from traits (*e.g.,* body size), phylogeny, co-occurrence, or theoretical constraints (Morales-Castilla *et al.*, 2015; Strydom *et al.*, 2021a). Network construction therefore constitutes a model-based inference step rather than a purely observational exercise. Different network construction frameworks encode distinct ecological assumptions about how interactions arise - whether as biologically feasible combinations of traits, energetically optimised realised diets, or topological structures. Note for the purpose of this work we will use the term 'model' to encompass the different approaches used to construct networks, acknowledging that the different frameworks are themselves different 'models'. These assumptions act as structural priors over network architecture (Petchey *et al.*, 2011; Guimarães, 2020; Gauzens *et al.*, 2025; Strydom *et al.*, 2026) and begs the question if alternative approaches systematically generate different trophic configurations. Risking conflating ecological differences among communities with

artefacts introduced by modelling choice. The reliability of macroecological inference therefore depends not only on ecological data, but on the structural assumptions embedded in the modelling approach.

Despite rapid methodological development in interaction inference, few studies have directly evaluated how alternative approaches to constructing networks influence macroecological conclusions when applied to the same species pool. This gap is particularly consequential for comparative research, where network metrics are routinely interpreted as indicators of environmental filtering, disturbance intensity, evolutionary history, or community stability (Allesina & Tang, 2012; Poisot *et al.*, 2015; Delmas *et al.*, 2018) and begs the question if models encode distinct structural constraints over interaction topology, then differences among communities may reflect model assumptions rather than ecological processes.

Deep time ecosystems provide an especially useful test of this issue because trophic interactions are not observed directly and must be inferred from traits, co-occurrence, and theoretical constraints (Roopnarine, 2006; Dunne *et al.*, 2008; Dunne *et al.*, 2014; Dunhill *et al.*, 2024; Smith *et al.*, 2025). As a result, ecological networks in these systems are inherently model-dependent, rendering the assumptions embedded within different models both explicit and consequential. Among such systems, the Early Toarcian Extinction Event (ETEE; ~183 Ma) provides a particularly informative case study. The ETEE was a major Early Jurassic biotic crisis associated with rapid climatic warming, widespread marine oxygen depletion, and substantial ecological turnover (Kemp *et al.*, 2024). Although considered a second-order extinction event globally, it nevertheless resulted in the loss of approximately 26% of marine genera worldwide (Little & Benton, 1995). Ecological disruption was especially severe in shallow marine ecosystems, where environmental stress drove extensive restructuring of benthic communities and food webs. In the Cleveland Basin of Yorkshire, the ETEE caused the extinction of around 60% of marine species, including up to 87% of benthic taxa (Caswell *et al.*, 2009). Fossil assemblages from this interval therefore record both the collapse of established communities and the subsequent recovery and reassembly of marine ecosystems under rapidly changing environmental conditions. Recent work has shown that the event selectively impacted specialist and infaunal organisms, leading to simplified ecosystems increasingly dominated by ecological generalists and opportunistic taxa (Dunhill *et al.*, 2024). These changes were accompanied by major shifts in community structure, trophic interactions, and ecosystem connectivity, with evidence suggesting that ecological recovery following the extinction was both prolonged and uneven

across marine habitats.

Ecological networks have previously been used to infer patterns of community organisation and extinction dynamics across this transition, providing an ideal test case for evaluating the robustness of network-based inferences of community response. Recent work by Dunhill *et al.* (2024) has used reconstructed food webs to draw ecological conclusions about community stability and collapse during this event. This creates a rare opportunity to ask a more general question - to what extent are such inferences contingent on the specific model used to generate the network?

Here, we reconstruct ecological networks for four successive assemblages across the ETEE using six contrasting modelling frameworks spanning feasibility-based, realised, and structural models. Holding species composition constant while varying only the model allows us to explicitly evaluate whether conclusions drawn from a single model (such as those of Dunhill *et al.* (2024)) are robust to alternative, equally plausible representations of interaction structure, or whether different models would lead to fundamentally different ecological inferences. More broadly, our study tests whether macroecological conclusions derived from ecological networks reflect underlying biological processes or are conditioned by the assumptions embedded within the modelling framework itself.

2 Methods

2.1 Study system and fossil data

We used fossil occurrence data from the Cleveland Basin, UK, spanning the late Pliensbachian to the late Toarcian. This interval encompasses a major volcanic-driven hyperthermal and marine extinction event. To capture network dynamics across this transition, we defined four successive palaeo-assemblages: pre-extinction (Pliensbachian), post-extinction (early Toarcian), early recovery (early–middle Toarcian) and late recovery (late Toarcian). Each taxon was characterised using their size and Bambach's ecospace framework (Bambach *et al.*, 2007), coding for tiering, motility, and feeding mode as per Dunhill *et al.* (2024). Each assemblage was treated as a community of potentially interacting taxa. The dataset includes 57 taxa across diverse groups (*e.g.,* cephalopods, bivalves, and gastropods). See Dunhill *et al.* (2024) for detailed descriptions of the dataset, including taxonomic composition, trait coding, and temporal resolution.

2.2 Network construction models

2.2.1 Conceptual classification of network types

Ecological network reconstruction encompasses a range of approaches that differ in how trophic interactions are inferred and in the ecological assumptions they encode. These approaches can be broadly grouped into three classes: feasibility-based, realised, and structural models. Together, these classes represent distinct hypotheses about how interactions arise and constrain the space of possible food web structure (Strydom *et al.*, 2026).

Feasibility-based models infer the set of potential interactions among species based on trait compatibility or mechanistic rules. These approaches define a feasible interaction space by identifying which consumer–resource pairs are biologically possible, without specifying which interactions are realised in each community. In palaeoecological contexts, where direct observation of interactions is not possible, such models provide a biologically grounded, formalised representation, of expert knowledge based rules that can be used to infer interactions (*e.g.,* Fricke *et al.* (2022); Roopnarine (2006); Shaw *et al.* (2024)).

Realised models aim to approximate the subset of interactions that are expected to occur in practice. These approaches impose additional constraints (such as energetic optimisation, mechanical limits, or probabilistic niche structure) on top of feasibility, thereby generating networks that represent putative realised diets (Brose *et al.*, 2006; Schneider *et al.*, 2016). Although often parameterised using body size or related traits, realised models differ in their underlying ecological assumptions about how consumers select resources and how trophic interactions are structured.

Structural models, in contrast, do not attempt to reconstruct empirical trophic interactions from species-level data. Instead, they generate networks based on general topological constraints such as species richness, connectance, or trophic ordering (Allesina *et al.*, 2008). These models are species-agnostic and therefore cannot be interpreted as reconstructions of specific ecological communities. Rather, they have the potential to serve as 'null hypotheses', providing reference expectations against which the structure and dynamics of data-informed reconstructions can be evaluated.

In this study, we selected models to span these three conceptual classes (Table 1) to sample distinct mechanistic interpretations of trophic interactions rather than exhaustively include all available implementations. Specifically, we include: a feasibility-based model

(PFWIM), representing trait-constrained interactions; multiple realised models (ADBM, ATN, and Body-size ratio), capturing energetic optimisation, mechanical constraints, and probabilistic allometric structure, respectively; and structural models (Random (Erdős-Rényi model) and Niche), providing theoretical reference points. Notably some approaches share common inputs (*e.g.,* body mass for ADBM, ATN, and Body-size ratio), they encode fundamentally different assumptions about how interactions arise. Additionally, although other feasibility-based approaches exist (*e.g.,* Roopnarine (2006)), they are conceptually similar to PFWIM in that they infer interactions from biologically feasible consumer–resource combinations. We therefore selected PFWIM as a representative feasibility-based framework because it formalises these rules in a transferable model that can be parameterised across datasets, rather than relying on system-specific expert assignment of links.

Table 1: Six different network modelling approaches that can be used to construct food webs for both this specific community but are also broadly suited to palaeo network prediction. These models span all facets of the network representation space (feasibility, realised, and structural network).

Model	Assumptions	Data needs	Limitation	Network type	Key reference	Usage examples
Random	Links assigned randomly	Species richness, number of links	Parameter assumptions, species agnostic	Structural	Erdős & Rényi (1959)	Null-model comparisons; testing whether observed network structure (connectance, motifs) deviates from random expectations
Niche	Species ordered along a 'niche axis'; interactions interval-constrained	Species richness, connectance	Parameter assumptions, species agnostic	Structural	Williams & Martinez (2008)	Evaluating trophic hierarchy and motif structure; baseline structural predictions

Model	Assumptions	Data needs	Limitation	Network type	Key reference	Usage examples
Allometric diet breadth model (ADBM)	Energy-maximizing predator diets	Body mass, abundance/biomass	Assumes optimal foraging; does not account for forbidden links	Realised	Petchey *et al.* (2008)	Predicting realized predator diets; exploring secondary extinctions
Allometric trophic network (ATN)	Links constrained by body-size ratios and functional response	Body mass, number of basal species	Assumes only mechanical/energetic constraints	Realised	Brose *et al.* (2006); Gauzens *et al.* (2023)	Simulating species loss; evaluating network collapse dynamics
Paleo food web inference model (PFWIM)	Interactions inferred using trait-based mechanistic rules	Feeding traits	Assumes feeding mechanisms; trait resolution required	Feasibility	Shaw *et al.* (2024)	Mapping feasible trophic interactions; assessing secondary extinctions

Model	Assumptions	Data needs	Limitation	Network type	Key reference	Usage examples
Body-size ratio model	Probabilistic assignment of links based on predator–prey size ratios	Body mass	Does not account for forbidden links	Realised	Rohr *et al.* (2010)	Estimating likely interactions; simulating cascading effects.

This classification allows us to distinguish variation arising from ecological assumptions embedded in models from variation attributable to species composition. By comparing networks generated from the same species pool across these complementary model classes, we explicitly evaluate how different representations of interaction structure condition ecological inference.

2.2.2 Network generation and replication

We evaluated six network reconstruction models using the approaches listed in Table 1, expanded descriptions of model assumptions, parameterisation, and link-generation rules are provided in Supplementary Material S1, including full mathematical formulations and parameter definitions. For each community, 100 stochastic network realisations were generated per model (n = 2400 in total, 600 per time bin). Of the simulated networks we excluded those that had a network metric that was considered an outlier ($> 3\sigma$ from the mean). This resulted in the removal of 13 networks (of which 3 were ATN model networks and the rest Random model networks), typically due to very large trophic level values. Where models required species body mass or trait values, these were uniformly sampled within the different size classes. This was done by sampling from a truncated lognormal distribution, with the the upper and lower bounds of each size class serving as the boundaries. Alternative distributions (lognormal and uniform) were tested, and shown to have a minimal impact on topology (Supplementary Material S1; Figure S1). Random and Niche models were parameterised using connectance values drawn from an empirically realistic range (0.05–0.25; Curtsdotter *et al.* (2011)). For the Random model,

which requires link number explicitly, connectance (Co) was first converted to expected link number (L) using $L = Co \times S^2$, ensuring consistency in network density across all models. Richness (S) was determined by the number of species for each time bin. For the Body-size ratio model, we followed the approach of Yeakel *et al.* (2014) using only the body-mass scaling to determine links between species. Sensitivity of network structure to these parameter choices is evaluated in Supplementary Material S1. The PFWIM generates a deterministic feasible metaweb for a given trait configuration. To allow for statistical analyses we introduced variation by applying a downsampling procedure that randomly pruned links until the connectance of the network decreased by 0.1. This step allowed us to introduce statistical variation while still preserving most of the original metaweb structure. The choice of the magnitude of connectance change and its effects on network structure are discussed in Supplementary Material S1.

2.3 Network metrics and structural analyses

We quantified network structure using a suite of network metrics Table 2, capturing overall network properties, motif structure, and species-level variability. Differences among models were assessed using a multivariate analysis of variance (MANOVA), with model identity as a fixed factor and the full set of network metrics as response variables. Variance partitioning was further assessed using permutational multivariate analysis of variance (PERMANOVA). Pairwise interaction turnover was quantified using link-based β-diversity and was calculated following the framework of Poisot *et al.* (2012) among the four of the modelling frameworks (ADBM, ATN, body-size ratio, and PFWIM). Random and Niche models were excluded because they are species-agnostic models that generate networks matching broad topological properties (*e.g.,* connectance and degree distributions) rather than predicting biologically realistic species-specific interactions. Specifically, we looked at interaction rewiring among shared species (β_{OS}, see S1 for additional details on the decomposition of interaction -diversity and its implementation), which allows separation of differences arising from altered interaction identities among species common to both networks. Because all networks within a given assemblage are constructed from the same species pool, differences in interaction structure primarily reflect 'rewiring' of trophic links due to stochasticity in the model rather than species turnover. All calculations were performed for all model combinations within the same assemblage (time bin).

Table 2: Network properties used for analysis as well as a brief description of how they are derived.

Metric	Definition	Reference
Connectance	L/S^2, where S is the number of species and L the number of links	Williams & Martinez (2004)
Maximum trophic level	Prey-weighted trophic level averaged across taxa	Williams & Martinez (2004)
Trophic Index (NDTI)	Normalised difference between the number of S1 (linear chains) and S2 (omnivory) motifs $(S1 - S2)/(S1 + S2)$	Milo *et al.* (2002); Dansereau *et al.* (2024)
Competition index (NDCI)	Normalised difference between the number of S4 (exploitative competition) and S5 (apparent competition) motifs $(S4 - S5)/(S4 + S5)$	Milo *et al.* (2002); Dansereau *et al.* (2024)
Generality	Standard deviation of normalised generality of all species within network, normalised by L/S	Williams & Martinez (2000)
Vulnerability	Standard deviation of normalised vulnerability of all species within network, normalised by L/S	Williams & Martinez (2000)

2.4 Extinction simulations and their evaluation

Following Dunhill *et al.* (2024), we simulated species loss from pre-extinction networks under trait-based, network-position–based, and random removal scenarios. Species were deleted sequentially, with cascading secondary extinctions allowed to propagate. Simulated post-extinction states were compared to observed (*i.e.,* reconstructed from fossil occurrence

data) networks using mean absolute differences (MAD) of food web metrics (Table 2) and modified true skill statistics (TSS) calculated separately at the node level (species presence/absence) and link level (presence/absence of interactions between species pairs). Scenarios were ranked within each model based on MAD and TSS performance, and Kendall's rank correlation coefficient (τ) was used to quantify agreement in scenario ordering across models. Detailed implementation of extinction sequences and secondary extinction rules is provided in Supplementary Material S1.

2.5 Software and Reproducibility

Ecological network reconstruction and extraction of structural metrics were conducted in Julia v1.11.4 (Bezanson *et al.*, 2017) using the model implementations provided in Supplementary Material S1. All statistical analyses, including MANOVA, PERMANOVA, and post hoc comparisons, were performed in R v4.5.2 (R Core Team, 2024). The full analytical workflow, including data preprocessing, network generation, extinction simulations, and metric calculation, is fully reproducible from the archived codebase.

3 Results

We found that different models (even those that appeared structurally similar) yielded fundamentally different ecological inferences. Across the six models the inferred food web structure, species interactions, and extinction dynamics differed. Multivariate analyses showed pronounced separation among models in network structure space (Figure 1), with model explaining most of the variance in structural properties (Figure 2). Notably, we had a consistenly high rate of species turnover between models (Figure 3) and differnces in inferences around extinction dynamics (Figure 4), demonstrating that structural similarity does not guarantee similarity in species-level diets or inferred behaviour of the system.

Model substantially influenced inferred extinction dynamics. Temporal trajectories of network collapse, interaction loss, and motif reorganisation differed among approaches (Figure 2). Although node-level extinction rankings were often broadly consistent, link-level outcomes and extinction inferences were highly sensitive to model assumptions (Figure 4). Together, these results show that ecological inferences drawn from networks are dependant on the model used.

3.1 Network structure differs among models

Across modelling approache, network structure (network properties listed in Table 2) differed significantly (MANOVA, Pillai's trace = 2.79, approximate $F_{30,11880} = 502.1$, $p < 0.001$), indicating that the model choice systematically alters inferred food web topology. Linear discriminant analysis (LDA) identified two dominant axes of variation, explaining 95.2% of the differences between models. LDA axis one correlated with trophic index, vulnerability, and generality. LDA axis two correlated with competition index, generality, and connectance (Figure 1; Table S1, Figure S1). Broadly indicating that structural differences are driven by the way in which the models assign links to species as opposed to macro-level network properties such as maximum trophic level.

[Figure 1 about here.]

3.1.1 Variance partitioning of network structure

Permutational multivariate analysis of variance (PERMANOVA) revealed that modelling framework accounted for most of the variation in multivariate network structure ($R^2 = 0.597$, $p < 0.001$), whereas temporal turnover across extinction phases explained a near negligable proportion of variance ($R^2 = 0.001$, $p < 0.001$), alongside the model × time interaction ($R^2 = 0.03$, $p < 0.001$), indicating limited but significant time-dependent divergence among models. Thus, differences among models were more than an order of magnitude greater than differences that could be associated with ecological turnover through the extinction-recovery period, even though the Pliensbachian–Toarcian dataset was characterised with a significant community turnover.

To determine whether the dominance of the model reflected absolute mean shifts among time bins, we repeated the analysis after centring network metrics within each extinction phase. This procedure removes between-phase differences while retaining within-phase structural variation. Even after temporal bin-standardised centring, the model explained the majarity of the observed multivariate variance ($R^2 = 0.704$, $p < 0.001$). These results demonstrate that the influence of modelling approach is not driven by temporal mean differences but reflects intrinsic divergence among models in how ecological interactions are organised.

3.1.2 Statistical Drivers of Network Variation

To identify which specific structural properties drive the multivariate separation observed above, we partitioned variance at the level of individual network metrics. Results show that model had a significantly stronger influence on network structure than the ecological signal of species loss. In panel A of Figure 2 for certain network metrics models predict different responses across time (*e.g.,* Trophic Index), as well as differing magnitudes of change (*e.g.,* connectance). A two-way factorial ANOVA across all eight network metrics confirmed that model was the dominant driver of variance, with partial eta-squared values (η_p^2) exceeding 0.7 (Figure 2, panel B; Table S3), with the exception of maximum trophic level for which $\eta_p^2 = 0.1$. While time (pre to post extinction recovery) significantly altered network structure ($p < 0.001$), its relative importance remained secondary, typically explaining a smaller fraction of the strucutral variation. This is clear in panel B of Figure 2 where all metrics are within the bottom-right (model-dominated) triangle (below the dashed line), emphasising that model assumptions outweigh the ecological signal of extinction-driven changes. Furthermore, the high inter-model Coefficient of Variation (CV) observed for some metrics (Table S4, Figure S4) highlights a sensitivity. The properties that are influenced (albeit modestly) by time are also those upon which models disagree most profoundly (larger point size). This demonstrates that our understanding of structural food web collapse in the fossil record is highly contingent on the chosen model, particularly when examining complex trophic pathways beyond simple macro-scale properties like connectance.

[Figure 2 about here.]

3.1.3 Inferred pairwise interactions vary widely among models

Despite some models approaches showing similar network metrics (*e.g.,* ATN and ADBM, Figure 1), specific pairwise interactions often differed. Pairwise β_{OS} revealed that certain model pairs shared very few links (Figure 3). Perhaps most striking is that the two size-based models (ADBM, ATN) that were broadly similar in structure still showcase a high degree of interaction turnover, demonstrating that agreement in overall network structure does not guarantee agreement in species-level interactions. As the PFWIM is a metaweb model on would expect that is should have the greatest dissimilarity with the other body-size constrained models - indeed we see a high degree of turnover, hoever this is not notably higher than turnover between the body size models. Overall we see

that at the pairwise level we see that each model tends to have a distinct set of pairwise interaction from the other. Not the the the moderate degree of interation turuover for within models from the PFIM can be ascribed to the the downsampling of links, wheres for the body-size ration this turnover can be ascribed to the fact that interactions are assigned a probability and thus we would expect *some* random noise.

[Figure 3 about here.]

3.2 Model choice influences inferred extinction dynamics

To evaluate how model choice affects inferred extinction dynamics, we compared simulated post-extinction networks to observed networks using mean absolute differences (MAD) for network-level metrics and true skills statistics (TSS) for node- and link-level outcomes (Figure 4). Across models, MAD-based rankings were generally positively correlated (Kendall's τ 0.122 across the structural metrics), indicating weak but generally positive agreement on the relative importance of extinction mechanisms despite the substantial differences observed between modelled network structure. The exception to this is maximum trophic level where we see more disagreement between the model rankings of extinction mechanisms.

Node-level TSS rankings were similarly consistent across models (Kendall's τ 0.528), reflecting broadly comparable node-level removal sequences. In contrast, link-level outcomes were far more variable (Kendall's τ 0.143), highlighting that inferences about which interactions are lost or retained during collapse and recovery are highly model contingent. However we see that models tend to converge on similar species-level extinction patterns, whereas from link-level extinction processes the inferred pathways of interaction loss and cascading dynamics depend strongly on the model.

[Figure 4 about here.]

4 Discussion

4.1 Network reconstruction is not neutral: assumptions shape ecological theory

Food web reconstruction is often treated as a methodologically neutral step, with network properties subsequently interpreted as reflections of underlying ecological organisation. This assumes that reconstructed webs provide structurally comparable representations

of ecological communities, such that differences in connectance, trophic structure, motif composition, or robustness reflect biological variation rather than modelling assumptions.

This assumption is particularly important in deep-time palaeoecology, where interactions are rarely observed directly and must be inferred through modelling. Although this makes making assumptions unavoidable, it also renders ecological interpretations sensitive to model choice, creating a risk that methodological artefacts are misinterpreted as genuine macroevolutionary or palaeoecological signals. The same concern applies to contemporary food webs, particularly as the field increasingly adopts prediction-forward approaches to network construction (Strydom *et al.*, 2021b).

Our results demonstrate that model choice explains substantially more variation in food web topology than temporal turnover across extinction and recovery phases. Despite sharing an identical regional taxon pool, alternative models generated distinct structural signatures occupying non-overlapping regions of multivariate space (Figure 1). Model type remained the dominant driver of structural variation even after accounting for differences between phases, indicating that models impose distinct *structural priors* that shape emergent topology, species roles, and predictions of extinction dynamics. Network structure is therefore determined not solely by ecological communities, but jointly by ecological data and the assumptions used to infer interactions (Gauzens *et al.*, 2025; Strydom *et al.*, 2026).

Variance partitioning further showed that model choice dominated variation across individual network properties, suggesting that its influence extends across multiple dimensions of network organisation rather than being driven by a single metric. Although this may partly reflect the propagation of constraints on individual structural properties into emergent network characteristics (Poisot & Gravel, 2014; Michalska-Smith & Allesina, 2019), constraining connectance to be identical across all replicates retained structural differences between models, albeit to a lesser degree (Supplementary Material S2). Notably, the properties exhibiting the strongest temporal signals also showed the greatest inter-model disagreement, suggesting that temporal trends are hardest to resolve where models diverge most strongly (panel B, Figure 2).

Model disagreement was even more pronounced at the level of individual pairwise interactions. Models producing structurally similar networks often inferred substantially different sets of species-level interactions (Figure 3), suggesting that interaction predictions are particularly sensitive to model choice. This indicates that models can impose distinct rules

governing network structure without agreeing on the specific links between species pairs. Importantly, this comparison concerns agreement between predictions, not their accuracy.

This pattern is particularly evident in the ATN and ADBM models. Both showed substantial variation along LDA axis 2, strongly correlated with the competition index, but relatively little variation along axis 1, strongly correlated with the trophic index and vulnerability (Figure 1). Together with the limited variation in trophic index across extinction and recovery phases (panel A, Figure 2), this suggests that these models constrain aspects of trophic structure, including the balance of linear chains and omnivory, despite variation in competitive structure. However, their structural similarity does not translate into agreement on individual interactions (Figure 3). Thus, models shape not only the overall structure of inferred food webs but also the ecological interactions through which that structure is realised.

4.2 Node- vs link-based inference

Importantly, the sensitivity of extinction simulation rankings to network type was not uniform across levels of inference. Node-level extinction rankings were broadly consistent among models, whereas link-level outcomes were model contingent (TSS panel, Figure 4). This likely reflects the fact that, for node-specific specifications, primary extinction order is determined by species traits and is therefore consistent across network types. The similarity in TSS scores, which evaluate the identity of surviving species, further suggests that secondary extinctions were also broadly consistent. However, when primary extinction sequences are determined by link-level traits, which depend on the inferred network structure, we see model-specific differences. This suggests that broad species-level inferences may be robust across plausible interaction architectures, whereas conclusions about interaction loss, retention, and reorganisation depend more strongly on how interactions are inferred.

This distinction challenges a central ambition of food web ecology: using interaction structure to identify the mechanisms underlying stability and collapse. While broad patterns may be robust across models, mechanistic explanations are far less secure. Had Dunhill *et al.* (2024) used a model other than PFWIM, the inferred drivers of extinction and recovery may have differed substantially. If extinction mechanisms vary across equally plausible reconstructions, mechanistic narratives derived from a single inferred topology may overstate their precision. The apparent determinism of extinction cascades may

therefore reflect model-imposed structure as much as ecological inevitability.

4.3 Implications for network-based research

These findings have important implications for comparative network studies. Because different models encode different ecological assumptions, observed differences among networks may partly reflect model choice rather than ecological processes (Brimacombe *et al.*, 2023, 2024). Without explicit standardisation or sensitivity analysis, methodological heterogeneity can be mistaken for biological signal. Food web ecology has devoted substantial effort to understanding how topology shapes dynamics (Dunne *et al.*, 2002; Allesina & Tang, 2012), but comparatively less attention has been paid to how reconstruction methods shape topology. Our findings indicate that these questions cannot be separated. However, disagreement among models does not mean that any single approach is necessarily wrong (Petchey *et al.*, 2011; Stouffer, 2019). Rather, different frameworks emphasise different ecological constraints, such as trait compatibility, energetic optimisation, or topological regularity. The challenge is therefore to match the model to the question at hand.

As network approaches are increasingly used to investigate ecological organisation across latitudinal gradients (Tylianakis *et al.*, 2008), climate-driven changes (Gilman *et al.*, 2010), and stability under species loss (Hao *et al.*, 2025; Marjakangas *et al.*, 2025), we need to consider how model choice may influence these results. Studies frequently interpret variation in connectance, trophic height, interaction ( )-diversity, and robustness as evidence of ecological differentiation across regions or through time (e.g., Pellissier *et al.*, 2018; Trøjelsgaard & Olesen, 2016). Our results show that model choice can systematically alter inferred topology and disturbance dynamics even when species composition is held constant. Apparent differences in network structure across spatial or climatic gradients may therefore reflect model choice as much as underlying ecological processes. Comparative studies using different modelling appraoches should consequently standardise their methods or explicitly quantify the sensitivity of their conclusions to alternative network representations.

No single model is universally correct. Rather, each represents a distinct hypothesis about how ecological interactions are constrained (Petchey *et al.*, 2011; Strydom *et al.*, 2026). Modelling choice should therefore be viewed as part of hypothesis specification rather than a technical step preceding analysis. Researchers must align modelling approaches with their ecological questions, whether estimating potential interactions, realised diets, or broad structural properties, because each framework emphasises different aspects of

ecological organisation. Our results show that the consequences of these choices depend on the scale of inference. Some conclusions, such as those related to species traits proved to be more robust to model choice than those related to node-level properties. Network-based inference may therefore be robust at coarse ecological scales while remaining imprecise in its mechanistic detail. Recognising this asymmetry is essential for distinguishing robust ecological patterns from model-dependent artefacts.

A more mature modelling paradigm in food web ecology would treat network construction as a testable hypotheses, incorporate probabilistic link inference where possible, and routinely evaluate the sensitivity of ecological conclusions to alternative representations of interaction structure. Such an approach aligns with recent advances in probabilistic and ensemble network modelling and would strengthen the interpretability of network-based inference under global change (Poisot *et al.*, 2016; Banville *et al.*, 2025).

5 Conclusions

Ecological network reconstruction is not a neutral methodological step, but a theoretical choice that shapes ecological inference. Applying six contrasting models to an identical species pool, we show that model choice systematically influences inferred network structure, interaction identity, and disturbance dynamics. While broad patterns, such as relative species vulnerability, were comparatively robust, fine-scale interactions and extinction cascades were highly model contingent. The reliability of network-based inference is therefore scale dependent.

These findings challenge the assumption that reconstructed networks are directly comparable across systems and through time. Differences in network structure and dynamics may reflect model assumptions as much as underlying ecological processes. Reconstruction should therefore be treated as an explicit component of hypothesis specification, rather than a technical step preceding analysis.

No single model captures the full complexity of ecological organisation, nor are alternative approaches interchangeable. Matching models to inferential goals, standardising approaches across comparative studies, and testing conclusions across alternative network representations are essential to distinguish robust ecological patterns from model-dependent outcomes.

Data and Code Availability Statement: The empirical data, derived network datasets,

and all analysis scripts are archived in a public Zenodo repository (DOI: 10.5281/zenodo.21200527), which includes a complete README file describing software dependencies, execution order, and reproduction instructions.

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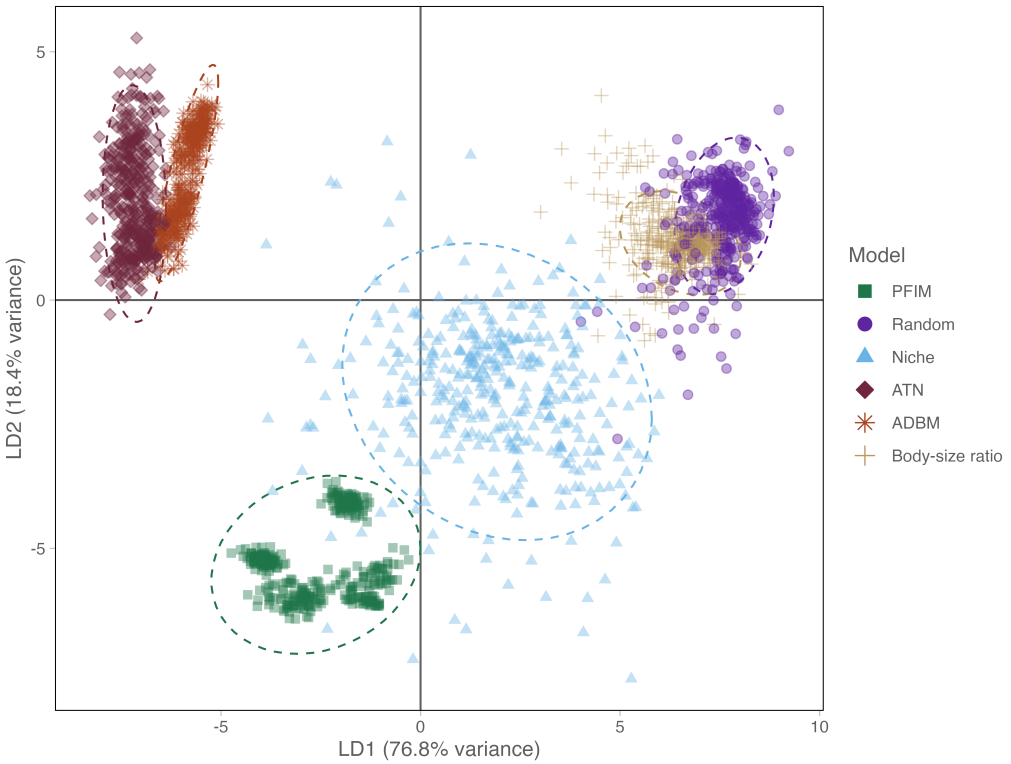


Figure 1: Linear discriminant analysis (LDA) of ecological network metrics for the six model types. Each point represents a replicate (n = 400 per approach), and ellipses indicate 95% confidence regions for each model.

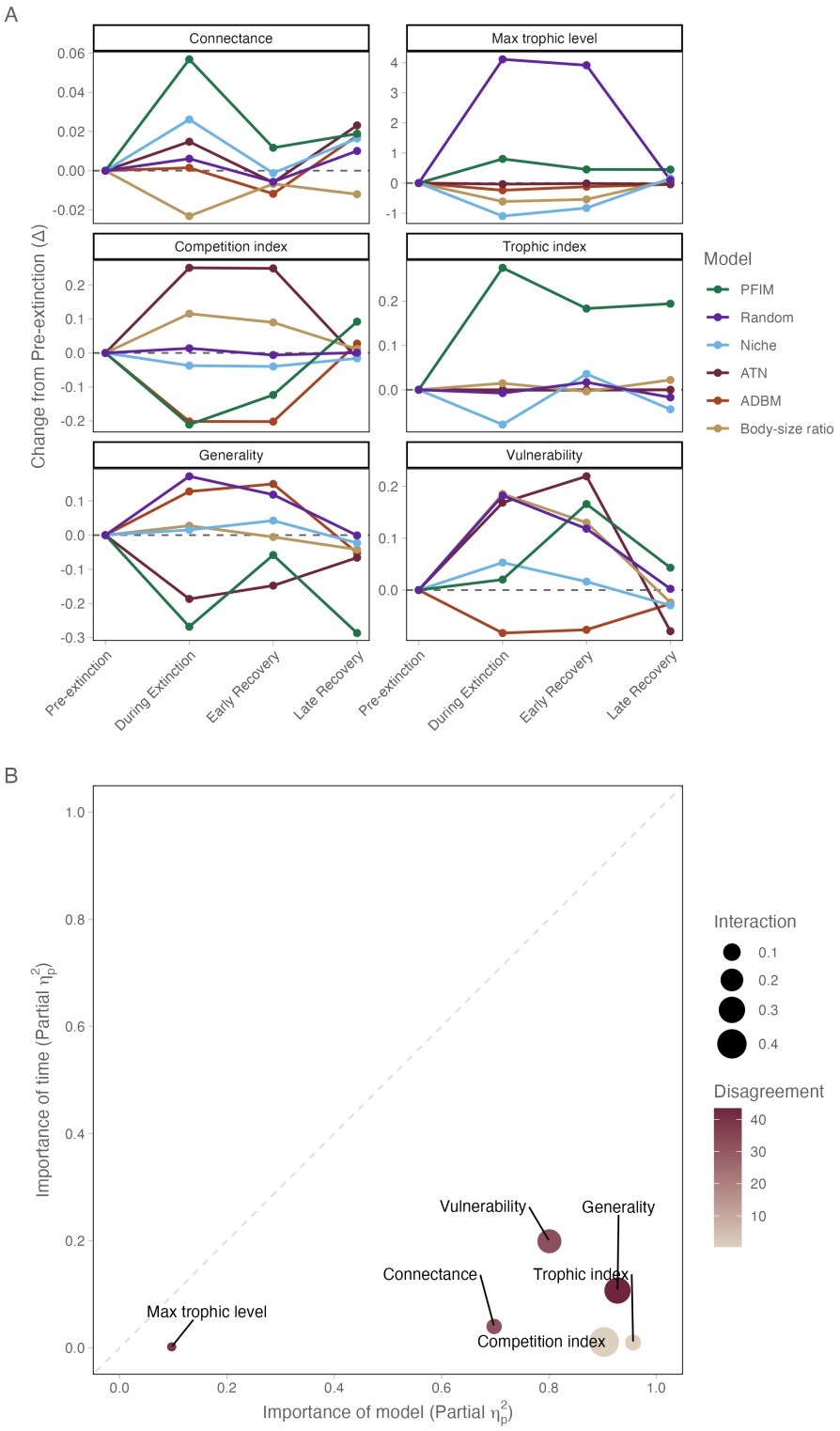


Figure 2: Temporal changes in food web structure and the relative influence of model and ecological turnover on network properties. (**A**) Changes in network metrics through time, expressed as deviations from pre-extinction conditions (δ). Metrics are faceted and scaled independently to highlight differences in temporal trajectories among network properties. (**B**) Relative influence of model (x-axis) and time (y-axis) on individual network metrics. The dashed 1:1 line indicates equal explanatory power. Metrics below the line are more strongly influenced by model than by temporal turnover. Bubble size represents the model × time interaction, and colour indicates disagreement among models (mean CV%), with darker colours indicating greater divergence in metric estimates.

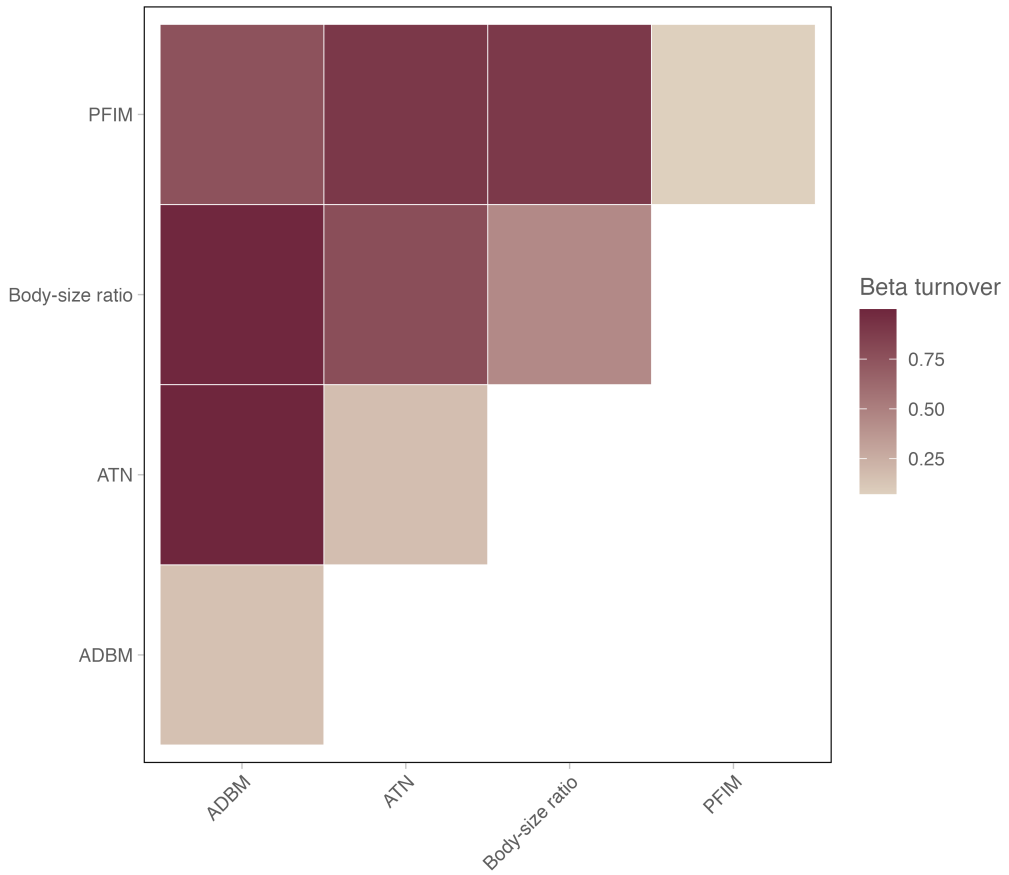


Figure 3: Pairwise interaction turnover (β_{OS}) among four food web models that infer species-specific trophic interactions (ADBM, ATN, Body-size Ratio, and PFWIM). Each cell represents the mean turnover value between a pair of models, with darker colours indicating greater dissimilarity in inferred interactions. High turnover values indicate strong disagreement in predicted trophic links, whereas lower values indicate greater similarity.

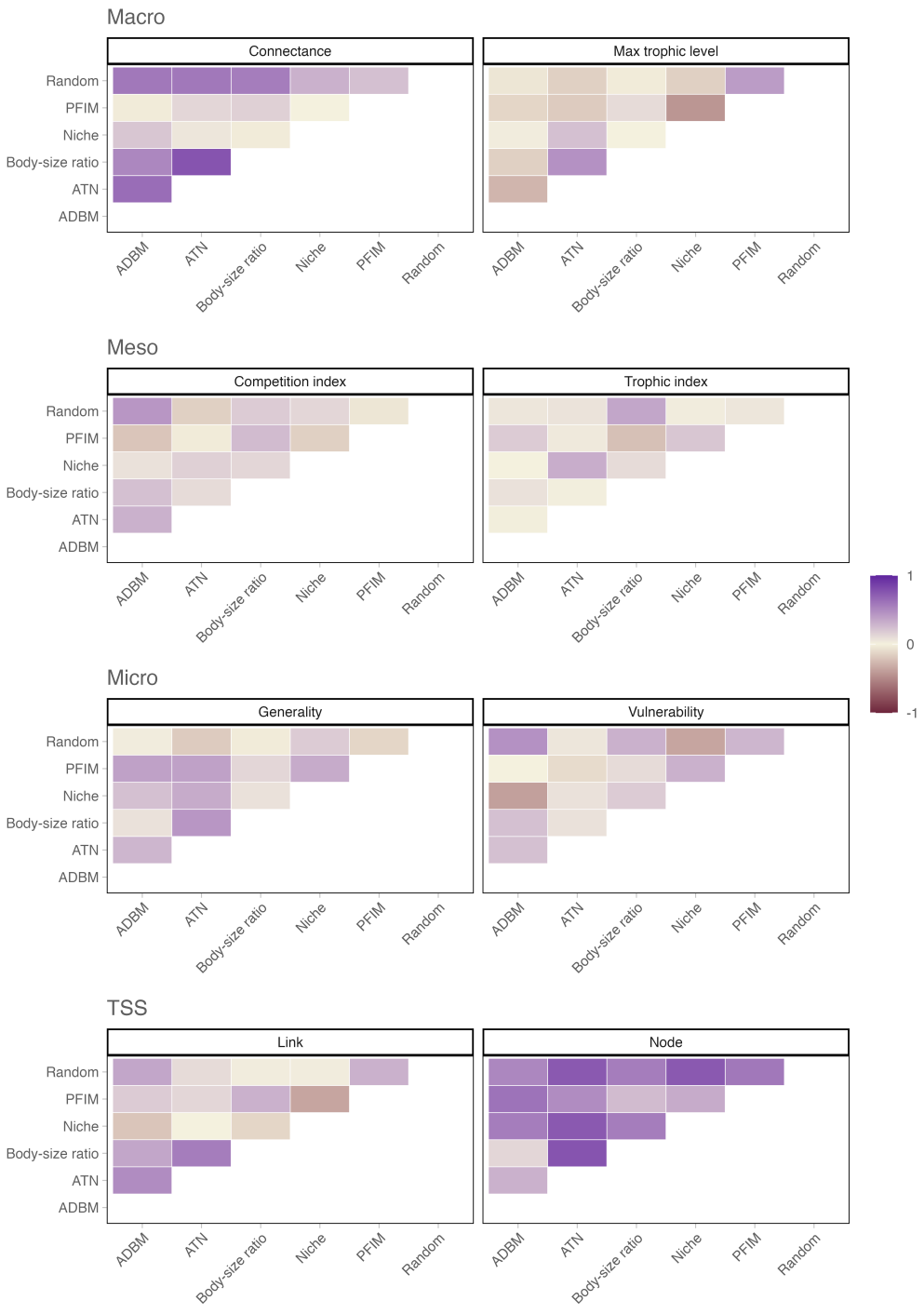


Figure 4: Heatmaps showing pairwise Kendall rank correlation coefficients (τ) between models for each network metric. Each panel corresponds to a different metric and displays the degree of agreement in extinction-scenario rankings across models based on mean absolute differences (MAD) between observed and predicted network values as well as differences in species links (TSS). Positive τ values (purple) indicate concordant rankings between models, whereas negative τ values (sienna) indicate opposing rankings. Paler colours approaching zero represent little or no agreement.